

SIGNificant Outlier Removal

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Abstract. Point clouds, usually obtained through scanning or various image processing, are commonly affected by noise and outliers. Such artifacts compromise data quality as they significantly distort subsequent processes, such as normal estimation and surface reconstruction. In this work, we introduce a proximity-based outlier removal method for point clouds. We improve on statistical methods based on neighboring graphs by using a parameter-free proximity graph - the spheres-of-influence (SIG), thus requiring fewer parameters compared to classical methods and obtaining better results. Moreover, the simplicity of our method allows it to become an easy replacement for existing statistical methods.

Keywords: Outlier Removal, Proximity Graphs, Point Clouds.

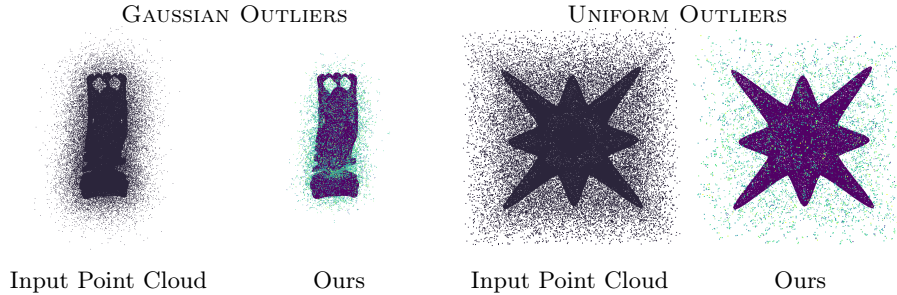


Fig. 1: Our method effectively deals with outliers of various kinds. The color of the points represents their distance to the ground truth CLOSE  FAR.

1 Introduction

Motivation Point clouds have become a prevalent form of data representation due to advancements in scanning technology. Yet, these unorganized points often require substantial processing to reconstruct the original surface or to derive other intrinsic characteristics. This procedure becomes even more complex considering that, usually, point clouds are corrupted with noise and outliers, either due to the scanning procedure, or the used reconstruction method if they are extracted from images. Outliers represent points that are not part of the surface but appear to be, due to various errors in the point cloud creation. Common issues

that lead to such errors might be reflective surfaces, semi-transparent surfaces, and otherwise inaccurate hardware. Noisy points, on the other hand, are part of the surface, but they are slightly shifted compared to ground truth due to similar issues. We, however, will focus on outlier removal, as our method improves on the statistical removal of outliers, while noise usually follows a more complex pattern and can rarely be removed by proximity-based methods alone - Figure 1.

Difficulties Numerous techniques for outlier removal have been developed, yet they frequently depend on the fine-tuning of multiple parameters specific to the features of the data. A common parameter in these methods is the *standard deviation ratio*, which determines how closely the data should cluster around the mean. However, merely adjusting this parameter is insufficient for robust outlier detection; a local neighborhood must also be defined, typically through the user-specified k in k -nearest neighbor (k NN) algorithms. Moreover, point cloud data often exhibits non-uniformity in sampling, further complicated by the presence of noise and artifacts from the acquisition process. These complexities render the task of parameter selection for scanned data both challenging and labor-intensive.

Contribution We propose a statistical outlier removal for point clouds, based on the *spheres-of-influence* graph (SIG), which is a parameter-free proximity graph. We evaluate our method on a diverse and recent dataset [24], that displays many of the common challenges faced by point clouds - noise and outliers, and we compare it with current state-of-the-art outlier removal methods, both statistical and data-driven, achieving better or comparable results to statistical methods. The main advantage of our method is its minimal parameter requirement, making it simpler to adapt to specific datasets. Due to this simplicity, our approach can readily replace traditional methods, offering a streamlined pipeline that yields superior results.

The main contributions of our work are, that the introduced method

- is an innovative statistical outlier removal method that leverages the spheres-of-influence graph,
- does not require extensive parameter tuning, simplifying the process and making it more accessible to users,
- and achieves results that are better or comparable to existing statistical methods.

2 Related Work

Outlier removal is a critical and complex domain that spans various branches of computer science, with particularly significant applications in the processing of point clouds. In this section, we provide a concise overview of the field, emphasizing techniques directly applicable to point clouds. Additionally, we discuss

relevant denoising methods, as many algorithms integrate both denoising and outlier removal to various degrees.

Several methods have been introduced in the wider context of outlier removal in various dimensions, with applications in multiple fields such as data mining, machine learning, or computer graphics. Approaches to outlier removal and denoising can be broadly categorized into several types. First, there are statistical methods, as highlighted by Aggarwal et al. [2]. Second, local filters, such as the bilateral filter adapted to point clouds [11] are commonly used, which consider normals for each point to enhance filtering accuracy. Another techniques involve projection methods [6,12,8,23], which typically approximate the surface using local planes based on the Moving-Least-Squares approach. These diverse strategies reflect the complexity and variety of challenges presented by the tasks of denoising and outlier removal in point clouds. The last two categories focus on denoising, while the last one provides various methods for surface reconstruction from noisy datasets. Hence, we discuss relevant classical methods in more detail.

Classical methods Classical methods typically identify outliers as isolated points distinguished by specific features like distance or the number of neighbors. These methods analyze data through probabilities, cluster formation, distances, or density variations. Extensive literature, including works by Aggarwal [2] and Rousseeuw [26], categorizes these approaches. We refer the reader to only some examples in each of the presented classes of methods, since providing a comprehensive list of works that use each method would be infeasible.

Probabilistic methods model the data as a distribution, given a specific set of requirements, and remove points that do not fit well the learned distribution [5]. The data can, for example, be modeled by Gaussian mixture models where parameters are derived using expectation maximization. These methods are advantageous due to their interpretability and their ability to provide nuanced insights based on statistical properties. However, they are sensitive to the choice of initial parameters and assumptions about data distribution, and can be computationally intensive, which limits their practicality.

Linear methods use the assumption that data is embedded in a lower dimensional subspace [4]. For example, a point cloud, which typically represents a physical surface, might be understood as a 2-dimensional subspace within a 3-dimensional space. This concept is utilized in methods such as Principal Component Analysis (PCA), which simplifies the complexity of data by identifying the main axes along which the data varies. PCA achieves this by calculating the eigenvectors and eigenvalues from the covariance matrix of the selected data subset. This reduction allows for the identification of outliers by examining deviations from these principal components. These deviations are often indicative of anomalies or noise in the data, making PCA a powerful tool for data cleaning and outlier detection.

Proximity-based methods take into account the distance between samples to extract outliers as isolated points. However, depending on how the locality of a point is defined, these methods can fall into one of the following categories:

cluster-based [3] - where a number of clusters is defined by the user and the points that do not strongly belong to any of the formed clusters are defined as outliers, *distance-based* [25,15] - where the average distance to the k -nearest neighbors is used to eliminate outliers whose distance is greater than a threshold, and *density-based* [7] - where a specific density is imposed around each point and if this is not reached, the point is considered an outlier and removed. Our method combines the advantages of distance- and density-based methods, as we do not only use distances for our computation but take the local density into account by working relative to the nearest neighbor. This approach encompasses more information about which points can be defined as outliers.

Machine Learning & Data Driven Approaches Recent advancements in point cloud processing prominently feature machine learning-based methods, particularly those utilizing end-to-end or self-supervised training frameworks. The two broad categories of data-driven approaches are discriminative and generative methods, each offering unique advantages in outlier removal and denoising tasks.

Discriminative Approaches. Usually, methods in this realm deal with the entire pipeline of outlier removal and denoising to obtain a completely clean point. Notable among these is PointNet [9] and its iteration, PCPNet [13], which incorporate a multiscale architecture to effectively manage local variations in the point clouds to learn local shape properties. PointCleanNet [24] exemplifies another widely used approach involving a two-stage denoising process. Initially, this method identifies outliers, and in the subsequent stage, adjusts individual points by predicting displacement vectors. This strategy contrasts traditional local surface fitting methods and provides improved handling of both dense and sparse regions within point clouds. Extensions, such as the PointProNets [27] architecture further improve on outlier detection. This fully differentiable network processes unordered input points by converting them into regularly sampled height maps. It effectively learns local parametrizations and fitted surfaces concurrently, preserving fine details.

Generative Approaches. Moreover, emerging approaches in point cloud denoising are increasingly exploring probabilistic and generative models. For instance, Luo and Hu [16] introduced a score-based method that adjusts the positions of points based on their statistical scores, aiming for improved detail preservation in complex geometries. Additionally, PD-Flow [17] incorporates normalizing flows to estimate displacement vectors within a point cloud’s gradient field. This approach provides a structured way to handle spatial distributions of points, ensuring that the adjustments adhere to the underlying data distribution. Such probabilistic models offer new avenues for enhancing robustness and fidelity in denoising complex point cloud datasets. However, even though data-driven approaches achieve impressive results in quality, the processing time is highly increased compared to classical methods, not only for training, which depends on the amount of data, but also for inference, which can take up to a few minutes for larger point clouds.

Spheres-of-influence The spheres-of-influence graph (SIG) has been introduced as a clustering method [28]. An edge is part of the SIG if the distance between its endpoints is less than the sum of the endpoints’ respective nearest neighbors. This graph has been previously used for estimating local densities for surface reconstruction [14] and used as an alternative to the classical k NN graph to encode connectivity in point clouds [20]. Moreover, the rarely used SIG has recently gotten attention in reconstruction methods [18,10,19] where combined with the Delaunay graph it showed promising results for curve and region reconstruction in the plane and on Riemannian manifolds.

3 Method

We define a point cloud as a collection of point P , situated in $\mathbb{R}^3$, containing individual points p , i.e. $P = \{p \in \mathbb{R}^3\}$. The set P , however, due to a variety of reasons that arise during the acquisition process - as discussed in Sec. 1, contains noise and outlier points as well, that do not belong to the surface. More precisely, the outliers are points that are situated far away from the surface and are therefore modeled as a separate set of points o tainting our actual inlier points p , i.e.,

$$P = \{p + n \in \mathbb{R}^3\} + \{o \in \mathbb{R}^3\}. \quad (1)$$

Our objective is to effectively remove outliers from a designated point set P . Typically, outliers are distinguishable by their significant distance from the main surface area, which is a distinct separation compared to merely noisy points within the dataset. To systematically address this, we first define a local neighborhood for each point p in the set P . This involves calculating the average length of edges connecting each point to its neighbors. This metric helps to assess the proximity and connectivity of points within the structure. Points whose average edge length is considerably different from the norm are considered outliers and are subsequently removed from the set. The process of identifying these local neighborhoods is facilitated by the spheres-of-influence graph, which connects each point to its immediate spatial neighbors based on geometric proximity.

Spheres-of-influence graph The SIG contains all the edges (v_0, v_1) such that for points v_0 and v_1 , the following holds:

$$\|v_0, v_1\| \leq nn(v_0) + nn(v_1), \quad (2)$$

Where $\|a, b\|$ denotes the Euclidean distance between points a and b , and $nn(a)$ signifies the distance from a to its nearest neighbor. This condition can be visualized by placing a ball at each vertex with a radius equal to the nearest neighbor distance for that vertex. Points are then connected if their respective balls intersect, as illustrated in Figure 2. This graph effectively represents the spatial proximity of vertices without requiring an additional parameter. It inherently includes the Nearest Neighbor graph [28], since the edge between a point and its nearest neighbor will always meet this criterion.

Since the distance to the nearest neighbor is implicitly included in the connection criterion, the graph's definition includes an approximation of the local density (local and global density might differ drastically; using local density through SIG allows for density-specific filtering). We assume that outliers appear as isolated points, where the nearest neighbor will be situated far away compared to the average edge length in the graph. The spheres of influence of outliers will then be large, connecting them to multiple points that are far away. Hence, we want to label as outliers the points whose average incident length is statistically larger than the average edge length.

Algorithm 1: SIGNificant outlier removal

Includes the fast SIG computation [20].

Data: $V = \{v \in \mathbb{R}^3 = P + O\}$ - point set with outliers, r

Result: P - clean point set

$P = \{\}$;

FAST COMPUTATION OF SIG

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SIG =  $\{\}$ ;
create kd-tree  $KT$  of  $V$ ;
for  $v \in V$  do
     $\lfloor$  find  $nn(v)$  using  $KT$ ;
for  $v \in V$  do
     $N = KT.findNbrInRange(2nn(v))$ ;
    for  $u \in N$  do
        if  $\|u, v\| \leq nn(u) + nn(v)$  then
             $\lfloor$  SIG +=  $(u, v)$ ;

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OUTLIER REMOVAL

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for  $v \in V$  do
    sum = 0;
     $N = SIG.findNbrs()$ ;
    for  $u \in N$  do
         $\lfloor$  sum +=  $\|u, v\|$ ;
    sum /= degree( $v$ );
    avg_length[ $v$ ] = sum;
 $\mu, \sigma = computeMeanStdDev(avg\_length)$ ;
for  $v \in V$  do
    if  $|avg\_length[v] - \mu| \leq r \cdot \sigma$  then
         $\lfloor$  P +=  $\{v\}$ ;

```

We aim to compute the SIG efficiently by utilizing the established theoretical distance bounds for SIG neighbors, as discussed in [20], along with their fast SIG algorithm. For each node, we calculate the average length of its one-ring connections and assign this average to each point in our dataset. Subsequently, we

compute both the median and the standard deviation of these average lengths. The algorithm requires a single parameter: r , a multiplier for the standard deviation (σ). This parameter determines the threshold for outlier removal, where points are excluded if their average neighbor length exceeds $r \cdot \sigma$. The complete process, including the fast SIG computation, is detailed in Algorithm 1.

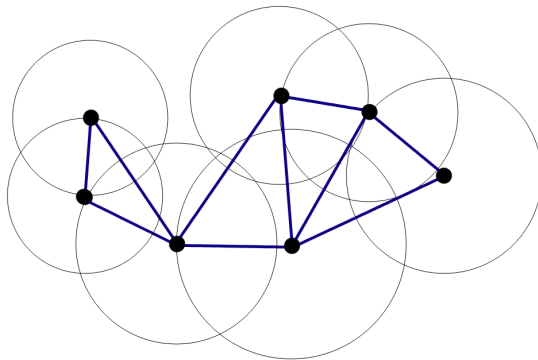


Fig. 2: Visual representation of the SIG connectivity in $\mathbb{R}^2$. We center a circle at each point with radius equal to the distance to its nearest neighbor. The intersections of circles generate the set of SIG edges [20].

4 Results

We compare our method to other outlier removal techniques for point clouds. However, since our method deals with outlier removal only and does not mitigate against noise, we cannot evaluate algorithms that deal with both noise and outliers, since that would not constitute a fair comparison. Hence, we compare our method to two classical, statistical methods that are most similar to ours: i) the statistical outlier removal (SOR) [22] filter, and ii) the radius outlier removal (ROR) [21] filter. Both classical methods are implemented in Open3D [1]. We also compare to the machine learning approach of PointCleanNet [24] in which we only consider the first stage that deals with outlier removal. While SOR classifies as outliers the points that are farther away from their neighbors than the average over the entire point cloud, ROR requires a minimum number of neighbors in a given radius, and removes those points that do not satisfy this requirement.

Parameters The classical methods require parameters to be chosen in advance. For SOR the parameters are the number of neighbors, and the threshold for average neighbor distance as a multiplier of the standard deviation of average neighbor lengths for the entire point cloud. For ROR, conceptually similar parameters need to be chosen, namely the number of neighbors that each point has to have, and the radius around each point in which these neighbors have to exist.

We vary the number of nearest neighbors among 5, 15, and 30, and the ratio among 1, 2 and 3. For the radius parameter, we use a percentage of the length of the bounding box diagonal. We chose the radius as 0.5%, 1%, or 2,5% of the bounding box diagonal. We use the pre-trained weights for PointCleanNet, which were obtained by using the dataset the same paper introduces [24].

Dataset We test on the dataset from PointCleanNet, which consists of 10 shapes, each of which is uniformly sampled with 140k points, to which two types of outliers are added. A random subset of 30% of the points is corrupted with Gaussian noise with a standard deviation of 20% of the bounding box diagonal. The second modality of outliers uses uniformly distributed noise inside a scaled-up bounding box by 10%. The starting meshes are either clean, or corrupted with 3 different levels of artificial white noise before the outliers are added. Hence, the testing set contains a total of $10 \times 2 \times 4 = 80$ point clouds.

Quantitative results Since we have access to the ground truth labels of the dataset, we chose to compare all the evaluated methods using the F_β score, see Equation 3. The F_β score computes a weighted combination of precision (real positives among all positive predictions) and recall (predicted positives among all real positives), measuring the success of a classification result. The weighting factor β allows for a trade-off between precision and recall, where $\beta = 1$ gives equal weight to the two measurements, and $\beta > 1$ gives more weight to recall. The F_β score is computed using the following formula:

$$F_\beta = (1 + \beta^2) \times \left(\frac{Precision \times Recall}{\beta^2 \times Precision + Recall} \right). \quad (3)$$

Similar to PointCleanNet, we use the F_1 and F_2 scores. We provide the best F -score for each input shape among all parameter variations, averaged over each noise category. We divide the dataset into inputs with outliers that follow a Gaussian, and Uniform distribution - Figure 3 and Figure 4 respectively. PointCleanNet achieves the best results for both F -scores, but does so at a high computational cost - Figure 7, taking 40 times longer than classical methods. Due to this mismatch of computational resources we will focus on comparing our method to the other two widely used classical methods.

For Gaussian outliers, we obtain the best F_1 -scores, which decrease once the noise level increases. This is to be expected, since when noise is introduced, there is no clear distance- or density-based boundary between inliers and outliers, which affects the performance of our method. We obtain a better score when precision and recall have equal weight, indicating a good classification of outliers in general.

When outliers are uniformly distributed in a scaled-up bounding box, we obtain the best F -score results among classical methods, except for the noisiest case for F_2 . Due to the outliers' distribution, our method is able to easily remove them, since their distances to neighbors and local densities are different compared to inliers.

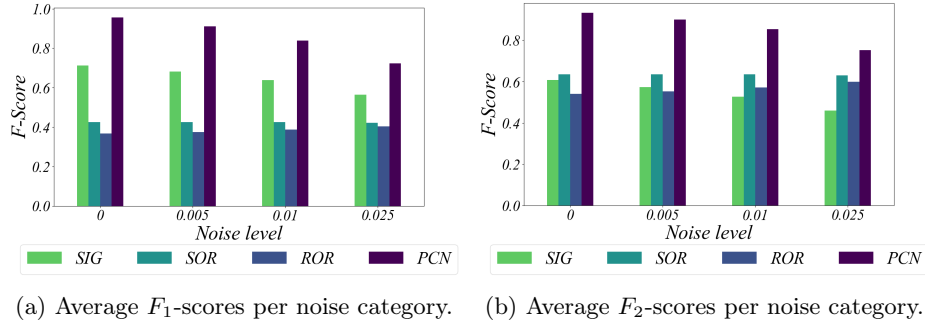


Fig. 3: F -scores for inputs with Gaussian outliers. Our method achieves the best results among classical methods for F_1 .

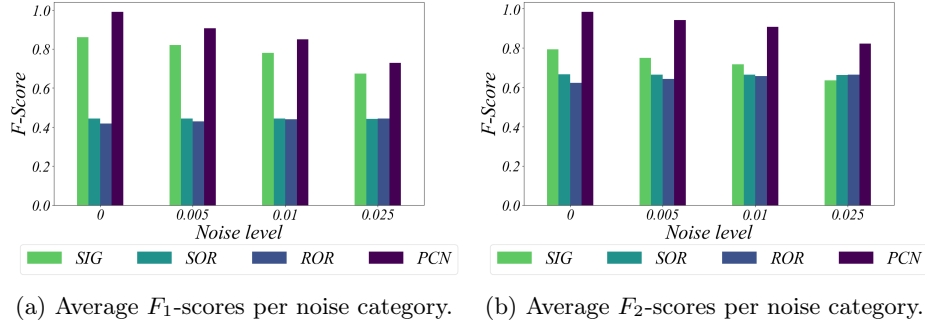


Fig. 4: F -scores for inputs with outliers following a uniform distribution. Our method achieves the best results among classical methods.

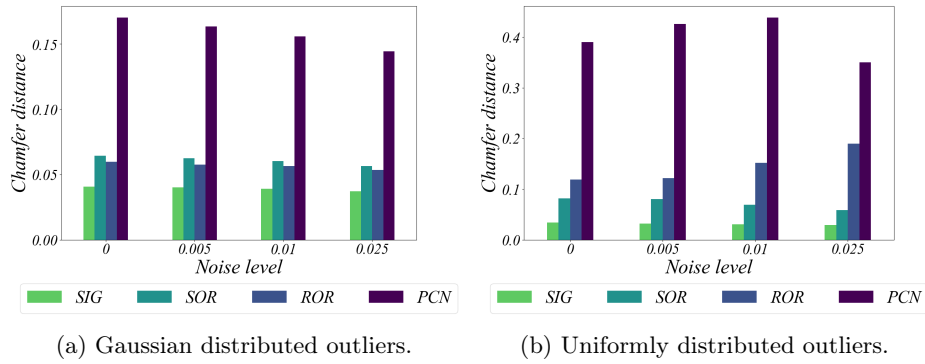


Fig. 5: Chamfer distance plotted for the evaluated methods between the ground truth point cloud and each method's cleaned results.

Besides the use of F -scores to evaluate our method’s results, we analyze the chamfer distance between the ground truth point set, and the cleaned point set obtained by filtering all points labeled as outliers by the various algorithms. We measure the symmetric minimum distance between ground truth and the resulting point cloud which illustrates how geometrically close the two point clouds are - Equation 4. We use the following formulation of the chamfer distance:

$$Chamfer(P, GT) = \frac{1}{|P|} \sum_{p \in P} \min_{q \in GT} \|p - q\|^2 + \frac{1}{|GT|} \sum_{q \in GT} \min_{p \in P} \|q - p\|^2, \quad (4)$$

where P is the cleaned point set with outliers removed, and GT is the ground truth point set without outliers.

All the point clouds in PointCleanNet are scale-normalized meaning that they fit into a unit bounding box diagonal. This makes the distances among the different shapes in the set comparable between each other.

In contrast to F -scores, which rely solely on correct or incorrect labels, we also assess the proximity of the cleaned point cloud to the ground truth. The results, illustrated in Figure 5, show that our algorithm consistently achieves the smallest distances to the ground truth point clouds. This is because our method incorporates local distances and density approximations for points by considering the nearest neighbor distance. This effectively removes outliers that are typically farthest from the ground truth surface and located in low-density regions.

Qualitative Results We present some examples of point clouds corrupted with outliers and our results in Figure 1, while Figure 6 illustrates the results of cleaned point clouds with the various methods we evaluated. We can observe that our results are similar to the other classical methods, validating our quantitative evaluation. The distance between the ground truth and the resulting point cloud is encoded in the color.

Timings We compare the computational cost of our method to the other evaluated algorithms, and we present the results in Figure 7. We excluded the PointCleanNet timings, since on average, each input takes about 320s to execute, which makes it 40 times slower than the classical methods, despite the improved F -score. Since all inputs in the PCN testing dataset have the same number of vertices (140k), we chose to illustrate the timing results in relation to the parameters that need to be chosen: k - the number of nearest neighbors and r - the ratio multiplier. Note that for the k -plot, the SIG results are constant, since this parameter does not influence our computation. We mention that Open3D’s SOR and ROR methods are highly optimized and parallelized, while our method is not. Overall, our method obtains results in a few seconds for 140k points, much faster than PointCleanNet. Even if SOR and ROR achieve faster computation times, our F -score and chamfer distance results are better.

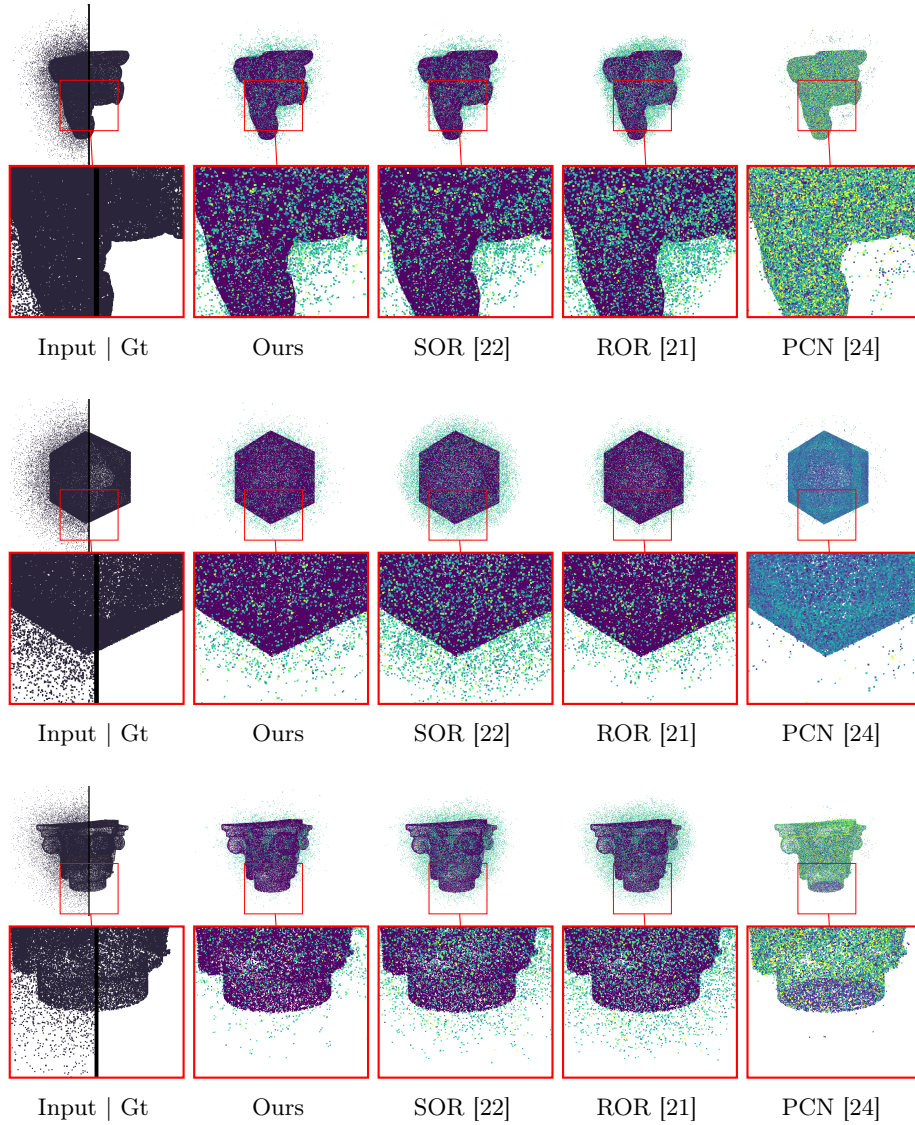
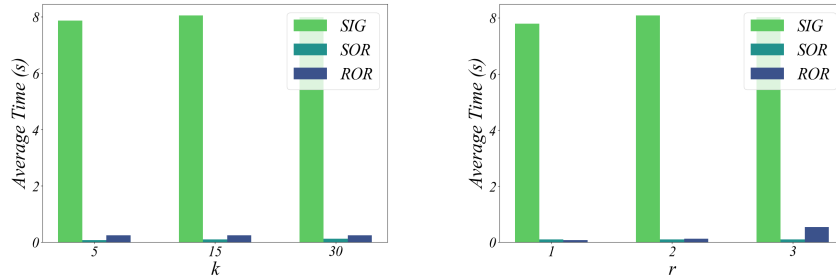


Fig. 6: Visual comparison of the evaluated methods. Our results are similar to SOR and ROR while only requiring one parameter. Points are colored based on their normalized distance to the ground truth surface. Since the distances are normalized per input set and PointCleanNet removes most of the outliers, the noise close to the ground truth is prominently highlighted, compared to the classical methods. The color of the points represents their distance to the ground truth with CLOSE  FAR.



(a) Varying the k parameter. SIG values are constant since this parameter does not influence our algorithm. (b) Varying the ratio r , Alg. 1 parameter. The ratios correspond to the diagonal percentages for ROR.

Fig. 7: Timings of evaluated methods while varying the input parameters. We showcase the average time taken per input shape. We excluded PointCleanNet [24] from the plots since, on average, each input takes about 320s to execute.

Limitations Due to the connectivity definition we use for our method, we are restricted to outliers that statistically differ from inliers by distance and density. If outliers follow a different, more complex distribution, our method is not able to correctly label all outliers, similarly to the other classical algorithms. Moreover, if the difference between outliers and noise is not significant in terms of distances to ground truth, our method struggles to differentiate the two. This is visible in Figure 6, where outliers close to the surface are not removed by any of the statistical methods.

5 Conclusions

We present an alternative to the commonly used k NN-based outlier removal methods, such as statistical outlier removal (SOR) and radius outlier removal (ROR). We introduce an algorithm that only requires a single parameter, and adapts to local distances and densities of point clouds. We use the spheres-of-influence graph to statistically extract outlier points whose average incident edge length differs significantly compared to other points. We have demonstrated that our method achieves results that are better or comparable to those of other classical approaches in classifying outliers. Additionally, the point clouds cleaned by our method more closely approximate the clean ground truth compared to those processed by all other methods, including data-driven ones.

Future work Our evaluation showcases that data-driven methods achieve better results than classical ones, and are also usually able to deal with noise, which usually accompanies outliers as the most common scanning artefacts. Hence, avenues of further development include investigating the possible applications of the spheres-of-influence graph to point cloud denoising. Combining our current

results with machine-learning-based methods, such as including our connectivity information in GNNs (Graph Neural Networks), could potentially yield significant improvements. Moreover, it would be interesting to investigate the applicability of our method to higher dimensions, especially in terms of optimization to counteract the curse of dimensionality.

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Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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